

```
def _find_cmd(cmd):
    #if not spawn.find_executable(cmd):
    if not shutil.which(cmd):
        print("*** %s not found: skipping" % cmd)
        return False
    return True
```

Figure 4: Removing *distutil* error

```
function install_mistral {
    setup_develop $MISTRAL_DIR
    # installing python-nose.
    if [ $OS == fedora ]; then
        if python3_enabled; then
            real_install_package python3-nose
        else
            real_install_package python2-nose
        fi
    else
        #real_install_package python-nose
        real_install_package python3-nose
    fi
    if [ $is_service_enabled horizon ]; then
        _install_mistraldashboard
    fi
    if [ "$MISTRAL_USE_MOD_WSGI" == "True" ]; then
        install_apache_wsgi
    fi
}
```

Figure 5: Removing 'python-nose' error

worldump.py file from 'distutil from spawn' to 'import shutil at line 22', and then remove the (line 79) 'spawn.find_executable' with 'shutil' (Figures 3 and 4).

The error occurs because the *distutil* library will be removed in Python 3.12.

If the error 'unable to install Python Nose' persists, it is because the system has not yet been configured to use Python as an alias for Python 3, and thus the code given in Figure 5 must be added. Go to 'opt/stack/mistral/devstack/plugin.sh' and update 'python-nose' to 'python3-nose' at line 138.

If there are errors with library dependencies, run '\$ sudo apt-get install aptitude' to help deprecate some libraries and avoid library inconsistencies.

```
This is your host IP address: 127.0.0.1
This is your host IPv6 address: ::1
Horizon is now available at http://127.0.0.1/dashboard
Keystone is serving at http://127.0.0.1/identity/
The default users are: admin and demo
The password: devstack

Services are running under systemd unit files.
For more information see:
https://docs.openstack.org/devstack/latest/systemd.html

DevStack Version: 2023.1
Change: 5e7afb779c469f593a1628e8f63c66989b7e2c49 Run dmesg command with sudo 2022-10-24 12:18:40 +0200
OS Version: Ubuntu 22.04 jammy

astha@astha-VirtualBox: ~/devstack$
```

Figure 6: Successful installation

Step 6: After all the errors have been rectified, run the *stack.sh* script again. This will install OpenStack with Tacker, and the terminal will show successful installation as in Figure 6.

Step 7: Launching the browser at *127.0.0.1/dashboard* will ask for authentication as shown in Figure 7.

On successful login, we get the dashboard with Tacker installed as shown in Figure 8.

The dashboard will show the networks, volumes attached, instances launched, etc.

Step 8: If you want to access the CLI of OpenStack in case you do not use the dashboard, download the OpenStack *rc* file as shown in Figure 9.

Step 9: Once you have successfully downloaded the *rc* password, which in my case is *devstack*, run OpenStack



The image shows the OpenStack login page. At the top is the OpenStack logo. Below it is a 'Log in' section with two input fields: 'User Name' with 'admin' entered, and 'Password' with a masked password '*****'. A 'Sign in' button is located at the bottom right of the login section.

Figure 7: OpenStack login authentication

commands from the terminal. Figure 10 shows the successful OpenStack authentication.

In this article, we have covered how to download and configure OpenStack on a machine. Troubleshooting steps were also

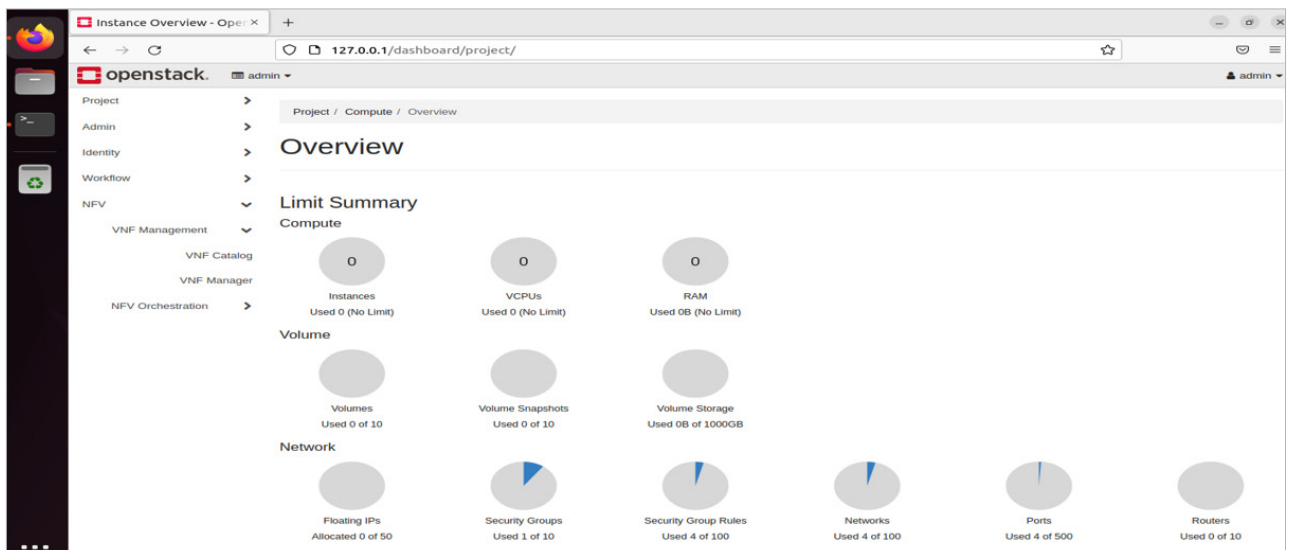


Figure 8: OpenStack dashboard

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